Project Title: The Role of Sialyltransferase ST6GAL1 in Breast Tumor Cell Phenotype

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Abstract

Glycosylation is an essential post-translational modification of proteins that dictates cellular functions. The sialyltransferase ST6GAL1-mediated α2,6-sialylation of glycoproteins has been implicated in regulating various events, including cell growth and survival, integrin-β1 function, inflammation, and cancer. ST6GAL1 resides within the ER-Golgi secretory apparatus of cells that produce them and is implicated in mediating its functions. ST6GAL-1 is prominently associated with human cancers, including breast cancers. However, many studies have found that enzymatically active ST6GAL1 is also released into the extracellular space. Extracellular ST6GAL1 has been shown as a potent modifier of hematopoiesis and in the sialylation of the anti-inflammatory IgG. However, the impact of this extrinsic mechanism of ST6GAL-1 on tumor cell pathobiology is unknown. In cancers, elevated ST6GAL1 generally predicts poorer patient outcomes, but the reason for these outcomes is unknown. We hypothesize that extracellular ST6GAL1 signaling is essential for aggressive triple-negative breast cancer (TNBC) cell growth and survival. We used recombinant ST6GAL1 (rST6G) for extracellular ST6GAL1 (exoST6G) protein and CMP-Sia, as the enzyme substrate for extracellular glycosylation mechanisms. Human TNBC MDA-MB-231 and BT549 cell models were used to examine extracellular ST6GAL1-mediated cancer cell biology. Our results show that the exoST6G mechanism enhances cancer cell α2,6 linked sialylation of glycoproteins. ExoST6G probably enhances tumor cell growth and survival by modulating cell-surface growth factor receptors, e.g., epidermal growth factor receptor (EGFR). In conclusion, our data suggested that extracellular ST6GAL1 signaling is important for cancer cell growth, survival, and invasion and is an easy target for therapy, which has strong clinical implications.